MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON, TYPE SOU-293

The complete requirements for procuring the tube described herein shall consist of this document and the latest issue of MIL-E-1.

DESCRIPTION: Two resonator klystron oscillator

ABSOLUTE RATINGS: Note 1

Parameters:	Ef	Ers	Irs	Ehk
Units:	Vdc	Vdc	mAdc	Vrms
Maximum:	6.9	1080	80	40
Minimum:	5.7	----	---	---
Test Condition: (Note 2)	6.3	Label voltage: Note 8	75 max.	---

		Ambient temperature		Tube body	
Parameters:	Frequency	Operating	Nonoperating	temperature	Load
Units:	mHz	°C.	°C.	°C.	VSWR
Maximum:	---	70	75	150	2:1
Minimum:	---	-30	-62	-62	---
Test Condition: (Note 2)	17,500 ± 5	24 ± 5	---	100 ± 10	1.1:1 max.

Weight: 12 ounces maximum.
Mounting position: Any.
Cooling: Forced air (see Note 2)
R.F. connections: Waveguide, mates with waveguide cover flange UG-419/U in accordance with MIL-F-3922/53

D.C. connections: AWG No. 20 molded leads (see figure 1 for lead connections).
Tuning: Fixed tuned.
Magnetic shielding: Required (see Note 12)
Ground: Anode grounded internally to tube shell.

GENERAL:

Qualification approval - Required

Holding period - Note 14

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Method	Requirement or Test	Conditions	Symbol	Limits		Unit
				Min.	Max.	
	Qualification					
---	Body temperature coefficient	Note 3	Δ F/ T	---	± 65	kHz/°C.
---	Ambient temperature coefficient	Note 4	Δ F/ T	---	± 10	kHz/°C.
---	Vibration	Note 5	Δ F(p-p) Δ Po	---	50 ±.5	kHz db
1042	Shock test	30G, 11ms half sine	Δ F	---	±1.0	MHz
---	Load variation	Note 15	- --	---		
	Quality conformance inspection, part 1					
1301	Heater current	Ef = 6.3	If	0.3	0.5	amps
4212	Resonator current	Note 14	Irs	---	75	mAdc
---	Label voltage	Note 8	Ers	950	1050	Vdc
4206	Emission	Notes 6 and 14	Δ Irs/Irs	---	10	%
4211	Power output	Notes 7, 8, and 12	Po	1.0	---	watts
---	A.C. bandwidth	Note 9	Δ F	15	---	MHz
	D.C. bandwidth	Note 10	Δ F	±2.5	---	MHz
	A.C. modulation sensitivity	Note 9	Δ F/ Δ Ers	---	250	kHz/v
	D.C. tuning rate	Note 10	Δ F/ Ers	---	150	kHz/v
	Quality conformance inspection, part 2					
4310	Pulling figure		Δ F	---	8	mHz
---	Amplitude modulation noise	Notes 11 and 12				
---	Dimensions	See figure 1				
	Quality conformance inspection, part 3					
---	Service life guarantee	Note 13	t	1000	---	hrs.

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NOTES:

1. The anode is grounded internally to the tube shell. Negative voltage is applied to the cathode with respect to ground. A minimum time interval of 1 minute is required between application of rated heater voltage and application of cathode voltage. These values should not be exceeded as tube life may be reduced. Absolute ratings apply to individual parameters. Attempts to operate at more than one maximum rating simultaneously may damage the tube.

2. The tube is designed to be cooled by forced air directed over the entire body. The flow may be interrupted in order to control the tube temperature. The tube body temperature is measured at point A shown on figure 1, by placing a #36, AWG wire thermocouple under the head of a #6-32 by 1/8 inch long binding head screw.

3. The body temperature coefficient is measured with a constant ambient air temperature. The tube shall be stabilized for at least 3 minutes at body temperatures of 110° and 80°C. The frequency shift shall be measured by means of a transfer oscillator and counter.

4. The ambient temperature coefficient is measured with a constant body temperature. The tube shall be stabilized for at least 3 minutes in ambient air temperatures of 60° and -20°C. The frequency shift shall be measured by means of a transfer oscillator and counter.

5. The tube shall be vibrated from 4 Hz (or the lowest attainable frequency) to 50 Hz, in discrete frequency intervals of 1 Hz, at the amplitudes shown in table I. at each frequency range, the vibration shall be maintained for 5 minutes. The frequency modulation and the change in output power resulting from vibration shall not exceed the limits specified.

Table I - Amplitudes of vibration

Frequency range (Hz)	Table amplitude ± inch
4 to 15	0.030 ± 0.006
16 to 25	.020 ± .004
26 to 33	.010 ± .002
34 to 40	.005 ± .001
41 to 50	.003 ± .000
	.001

6. Under conditions in Resonator Current Test, the heater voltage is decreased from 6.3 to 5.7 V. The percentage change in resonator current, after 1 minute.

$$\frac{\text{Irs (Ef = 6.3 V)} - \text{Irs (Ef = 5.7 V)}}{\text{Irs (Ef = 6.3 V)}} \times 100$$

shall be less than the maximum specified value and shall be defined as emission.

7. All oscillation tests shall be made with the tube rigidly connected to a UG-419/U Cover Flange in accordance with MIL-F-3922/53 on appropriate RG-91/U Waveguide in accordance with MIL-W-85.

8. The output power shall be measured at the mode-center frequency. The resonator voltage at the mode-center frequency shall be defined as the label voltage.

9. The A.C. bandwidth is measured between -0.5 db power points. The resonator voltage shall be modulated at a frequency in excess of 50 Hz. A.C. modulation sensitivity is the A.C. bandwidth divided by the peak to peak value of the modulating voltage.

10. The D.C. bandwidth is defined as the frequency deviation between -0.5 db power points obtained by variation of the D.C. resonator voltage. The D.C. tuning rate is the D.C. bandwidth divided by the D.C. voltage change required to produce the tuning.

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11. An appropriate sample of unmodulated klystron output shall be applied to a calibrated microwave detector which responds to amplitude only. A D.C. voltmeter shall be used to indicate carrier power. A low noise A.C. amplifier shall be used to isolate the A.C. measuring equipment. The output of this amplifier shall be applied to a filter and the output of the filter shall be measured with an rms voltmeter. The bandwidth of these filters and the ratio of power in this bandwidth to carrier power shall be as follows:

Filter bandwidth	Deviation from carrier	Maximum ratio
200 Hz	1 through 7 KHz	-100 db
1 KHz	10 through 100 KHz	-100 db

12. Adequate external magnetic shielding shall be provided by the user to prevent any reduction in output power or spurious modulation of the output signal due to the presence of magnetic materials or magnetic fields in close proximity to the tube.

13. End of life is defined as that point where the tube power output drops to less than 0.8 watt.

14. Resonator current and emission tests shall be performed after completion of the holding period.

15. The tube, with mode-shaping transformer attached, shall not be damaged by operation into any phase of a 2.0 to 1 maximum VSWR rf load.

User activity:
Navy-AS, Os, MC, CG, SH

Preparing activity:
Navy - EC

(Project 5960-N835)

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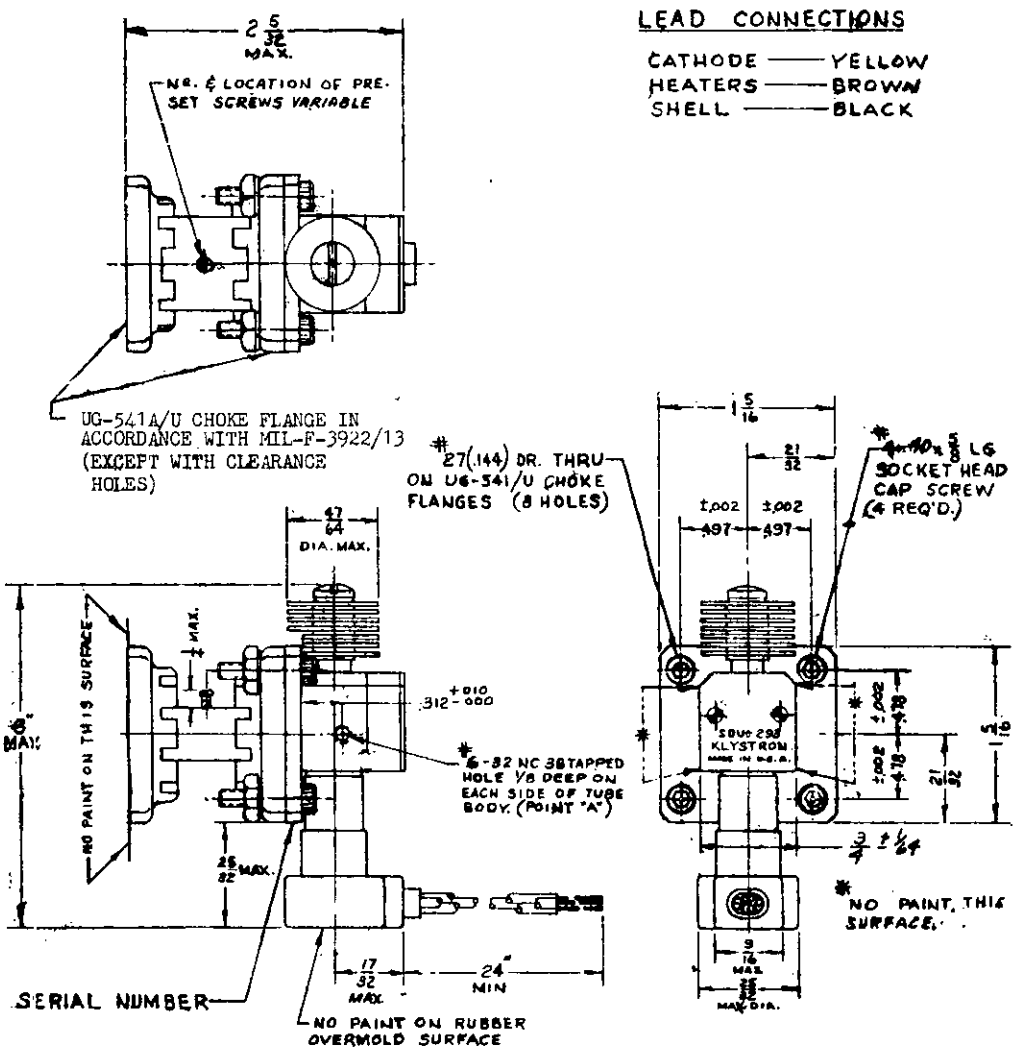


Figure 1 - Outline drawing for electron tube types SOU-293

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